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Beyond the Index Lesion: The Added Value of Breast MRI in Invasive Lobular Carcinoma — A Two-Case Series

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BACKGROUND

Invasive lobular carcinoma (ILC) is the second most common invasive breast malignancy, accounting for 5–15% of cases. Its infiltrative growth pattern may result in underestimation of disease extent on conventional imaging. Contrast-enhanced breast MRI has high sensitivity for ILC and plays an important role in local staging. We present two cases illustrating its added value in detecting additional ipsilateral and contralateral disease.

CASE PRESENTATION

Case 1: A 55-year-old woman presented with a right breast lump. Mammography and ultrasound demonstrated a highly suspicious right breast mass with satellite lesions and bilateral axillary lymphadenopathy. Biopsy of the right breast mass confirmed ILC, while the left axillary lymph node demonstrated metastatic carcinoma. Breast MRI (Figure 1) demonstrated multicentric right breast disease and an additional suspicious left breast lesion at 4 o'clock. This lesion was identified on second-look ultrasound, and ultrasound-guided biopsy yielded ILC, establishing bilateral breast malignancy. The patient was started on chemotherapy with palliative intent.

Case 2: A 44-year-old woman presented with a right breast lump. Mammography and ultrasound demonstrated a suspicious right breast lesion at 11 o'clock, with biopsy confirming ILC. Breast MRI (Figure 2) demonstrated the known malignant mass and an additional suspicious right breast lesion at 9 o'clock. This lesion was identified on second-look ultrasound, and ultrasound-guided biopsy confirmed ILC with lobular carcinoma in situ (LCIS). The patient underwent right wide local excision with bilateral reduction mammoplasty. Postoperative histopathology demonstrated multiple foci of ILC and LCIS in the right breast and left LCIS, confirming more extensive disease.

DISCUSSION

IILC can be challenging to detect and accurately stage on conventional imaging (mammography and ultrasound) due to its infiltrative, single-file growth pattern, often without formation of a discrete mass or significant desmoplastic response.

Unlike invasive ductal carcinoma, ILC may manifest as subtle architectural distortion or asymmetry on mammography and ultrasound, leading to underestimation of tumour extent. ILC may also be larger than anticipated and frequently presents with multifocal or multicentric disease.

Breast MRI offers high sensitivity for ILC and is particularly valuable in defining tumour extent, detecting additional ipsilateral and contralateral disease, and guiding surgical planning.

FIGURES

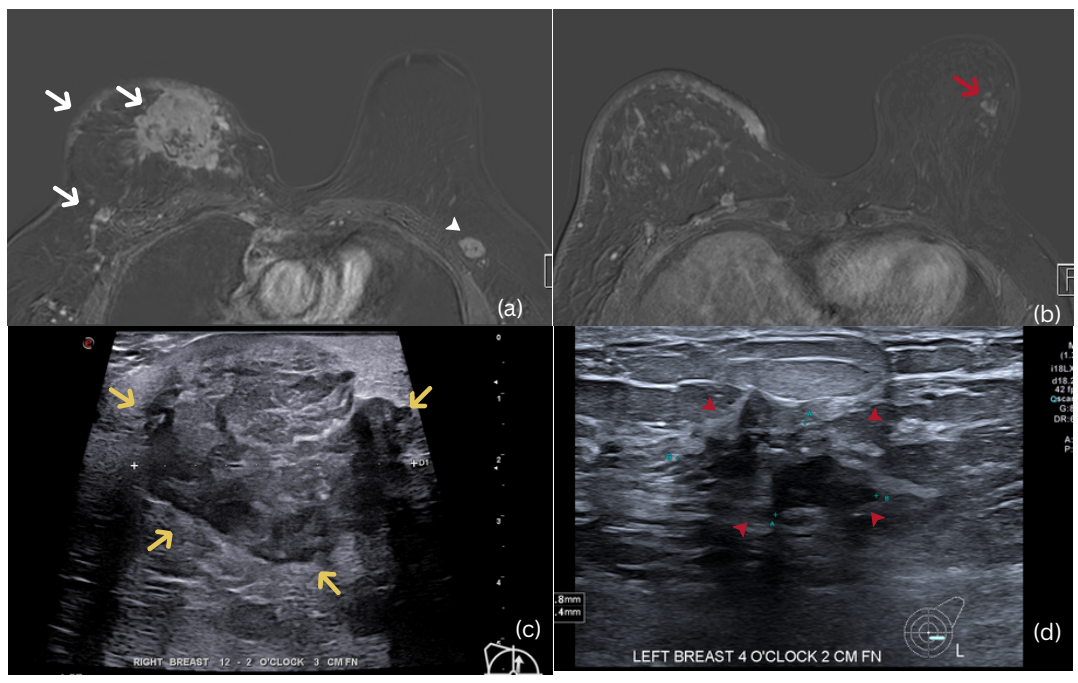

Figure 1 MRI findings (a) multicentric disease in the right breast with skin involvement (arrows). Additional findings of left enhancing axillary lymph node (arrowhead) (b) linear non mass enhancement at left 4:00 (red arrow). Ultrasound findings (c) malignant mass at right breast 12-2:00 3cmFN (yellow arrows). (d) irregular, indistinct, hypoechoic mass at left breast 4:00 2cmFN (red arrowheads).

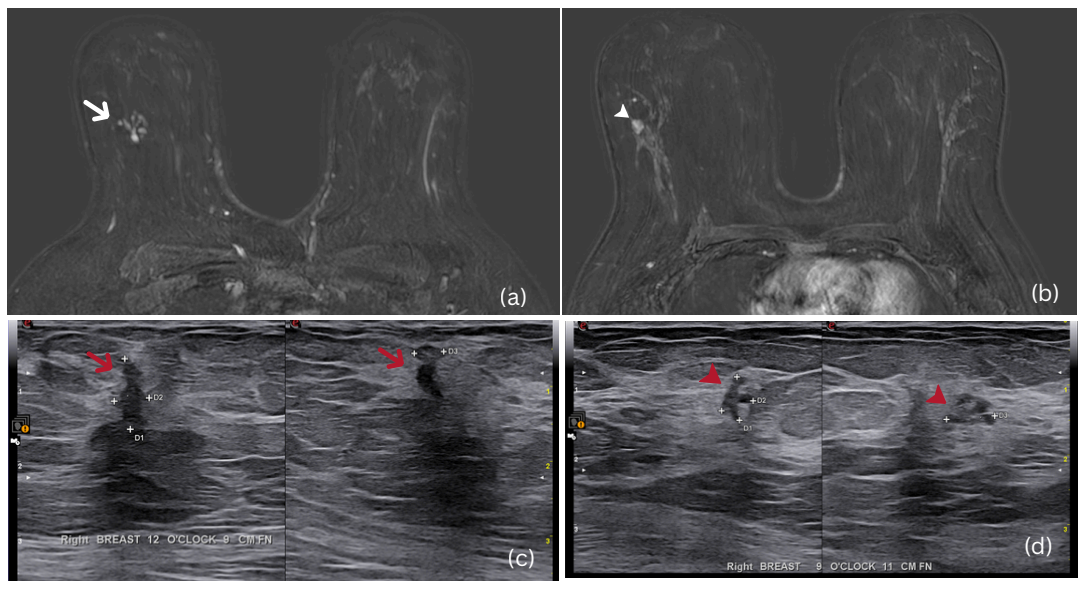

Figure 2 MRI findings (a) malignant mass at right breast 11:00 (arrow). (b) irregular, heterogeneously enhancing lesion at right breast 9:00 (arrowhead). Ultrasound findings (c) malignant mass at right breast 11:00 (red arrows) (d) irregular, indistinct, hypoechoic lesion at right breast 9:00 (red arrowheads).

CONCLUSION / TAKE-HOME MESSAGE

Breast MRI is essential in local staging of ILC, enabling detection of additional ipsilateral and contralateral disease and facilitating appropriate biopsy, staging, and individualized surgical planning.

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